

Worked examples for Lecture 23: Mode splitting and coupling

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These examples accompany Lecture 23. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Example 1 works through the toroidal mode problem for a homogeneous earth model in the detail referred to in the lecture summary. Numerical results were checked with the scripts in the python directory.

Example 1: Toroidal modes of a homogeneous sphere. In the lecture it was stated that, within an entirely solid and spherically symmetric earth model of radius b , the eigenvalue problem for a toroidal mode of degree l reduces to

$$-\omega^2 \int_0^b \rho W'^* W r^2 dr + \int_0^b \mu \left(r \frac{dW'^*}{dr} - W'^* \right) \left(r \frac{dW}{dr} - W \right) dr + (\zeta^2 - 2) \int_0^b \mu W'^* W dr = 0, \quad (1)$$

which is to hold for all test functions W' , where $\zeta^2 = l(l+1)$ and W is the radial eigenfunction. (i) By integrating by parts, show that eq. (1) is equivalent to an ordinary differential equation of Sturm–Liouville type for W on $0 < r < b$ together with a boundary condition at $r = b$, and interpret this boundary condition physically. (ii) Show that when ρ and μ are constant the solution regular at the centre is $W(r) = j_l(kr)$, with $k = \omega/\beta$, $\beta = \sqrt{\mu/\rho}$, and j_l the spherical Bessel function of the first kind, and obtain the equation that determines the eigenfrequencies. (iii) Solve this equation numerically for $l = 2$ and $l = 3$, tabulating the periods of the first three overtones for a homogeneous sphere with $b = 6371$ km and $\beta = 6.3$ km s⁻¹, and compare the period of ${}_0T_2$ with the observed value of 44.2 min.

Solution. Throughout this example there is no time dependence, and so we use a dot to denote d/dr ; as in the lecture, W' is the test function and not a derivative.

(i) *The radial equation and its boundary condition.* Expanding the product in the second integral of eq. (1) gives

$$\mu (r \dot{W}'^* - W'^*) (r \dot{W} - W) = \mu r^2 \dot{W}'^* \dot{W} - \mu r \dot{W}'^* W - \mu r W'^* \dot{W} + \mu W'^* W, \quad (2)$$

and we integrate by parts the two terms that contain the derivative of the test function,

$$\int_0^b \mu r^2 \dot{W}'^* \dot{W} \, dr = \left[\mu r^2 W'^* \dot{W} \right]_0^b - \int_0^b W'^* \frac{d}{dr} (\mu r^2 \dot{W}) \, dr, \quad (3)$$

$$- \int_0^b \mu r \dot{W}'^* W \, dr = - \left[\mu r W'^* W \right]_0^b + \int_0^b W'^* \frac{d}{dr} (\mu r W) \, dr. \quad (4)$$

The boundary terms at $r = 0$ vanish so long as W , $\dot{W}$ and W' remain bounded at the centre, which we assume. Since $\frac{d}{dr}(\mu r W) = r \dot{\mu} W + \mu W + \mu r \dot{W}$, the term $\mu r \dot{W}$ cancels against the third term on the right of eq. (2), and collecting everything eq. (1) becomes

$$\int_0^b W'^* \left[-\frac{d}{dr} (\mu r^2 \dot{W}) + (\zeta^2 \mu + r \dot{\mu}) W - \omega^2 \rho r^2 W \right] \, dr + b^2 W'^*(b) T(b) = 0, \quad (5)$$

where we have introduced the function

$$T(r) = \mu \left(\dot{W} - \frac{W}{r} \right) = \mu r \frac{d}{dr} \left(\frac{W}{r} \right). \quad (6)$$

Eq. (5) must hold for every test function. Considering first those that vanish at $r = b$ but are otherwise arbitrary, the square bracket must vanish for $0 < r < b$, and so W satisfies

$$\frac{d}{dr} \left(\mu r^2 \frac{dW}{dr} \right) - \left(\zeta^2 \mu + r \frac{d\mu}{dr} \right) W + \omega^2 \rho r^2 W = 0. \quad (7)$$

Given eq. (7), the integral in eq. (5) vanishes identically, and what remains must vanish for arbitrary values of $W'(b)$. Hence we also require

$$T(b) = \mu \left(\frac{dW}{dr} - \frac{W}{r} \right) \Big|_{r=b} = 0. \quad (8)$$

Conversely, if W satisfies eqs. (7) and (8) then eq. (5), and hence eq. (1), holds for every test function; the two formulations are equivalent.

Eq. (7) has the Sturm–Liouville form $\frac{d}{dr}(p\dot{W}) - qW + \lambda wW = 0$ with $p = \mu r^2$, $q = \zeta^2 \mu + r \dot{\mu}$, weight $w = \rho r^2$, and eigenvalue $\lambda = \omega^2$. Standard theory for such problems then gives directly what the lecture stated in general: the squared eigenfrequencies are real, eigenfunctions with distinct eigenfrequencies are orthogonal with respect to $\int_0^b \rho W_n W_{n'} r^2 \, dr$, which is just the kinetic energy form restricted to toroidal modes of degree l , and there is a countable infinity of eigenvalues accumulating at infinity, labelled by the overtone number $n = 0, 1, 2, \dots$. The boundary condition in eq. (8) was not imposed but emerged from the weak form; it is a *natural* boundary condition. Its physical meaning follows from computing the traction on a spherical surface. For a toroidal displacement $\mathbf{s} = W \mathbf{C}_{lm}$ the radial component vanishes, and the shear strains in an isotropic model give $\sigma_{r\theta} = \mu(\dot{W} - W/r)C_\theta$ and $\sigma_{r\phi} = \mu(\dot{W} - W/r)C_\phi$, so that the traction acting on the surface $r = \text{const}$ is $T(r) \mathbf{C}_{lm}$. Eq. (8) is therefore precisely the traction-free condition at the free surface, as was to be expected from the discussion in Lecture 22 of how boundary conditions are built into the weak form. It is also worth noting the identity $r\dot{W} - W = r^2 \frac{d}{dr}(W/r)$: the first potential energy term in eq. (1) vanishes for $W \propto r$, which for $l = 1$ is a rigid rotation, and we return to this point below.

(ii) *The homogeneous sphere.* If ρ and μ are constant then $\dot{\mu} = 0$, and dividing eq. (7) by μr^2 we obtain

$$\frac{d^2 W}{dr^2} + \frac{2}{r} \frac{dW}{dr} + \left[k^2 - \frac{l(l+1)}{r^2} \right] W = 0, \quad k = \frac{\omega}{\beta}, \quad \beta = \sqrt{\frac{\mu}{\rho}}. \quad (9)$$

In terms of $x = kr$ this is the spherical Bessel equation of order l , with general solution $W = A j_l(kr) + B y_l(kr)$. Near the centre, trying $W \propto r^\alpha$ in eq. (9) gives at leading order $\alpha(\alpha+1) = l(l+1)$, so that $\alpha = l$ or $\alpha = -(l+1)$; these are the behaviours $j_l(x) \sim x^l / (2l+1)!!$ and $y_l(x) \sim -(2l-1)!! / x^{l+1}$ of the two solutions. The second is unbounded, and would give an infinite kinetic energy since $\int_0 \rho W^2 r^2 dr$ then behaves like $\int_0 r^{-2l} dr$ for $l \geq 1$. Hence $B = 0$, and up to an arbitrary amplitude

$$W(r) = j_l(kr). \quad (10)$$

The eigenfrequencies follow from the traction-free condition in eq. (8), which reads $\mu[kj'_l(kb) - j_l(kb)/b] = 0$. Writing $x = kb = \omega b / \beta$, this is

$$xj'_l(x) = j_l(x), \quad \text{or equivalently} \quad \frac{d}{dx} \left[\frac{j_l(x)}{x} \right] = 0. \quad (11)$$

Using the recurrence relation $j'_l(x) = \frac{l}{x} j_l(x) - j_{l+1}(x)$ we can also write the frequency equation as

$$(l-1)j_l(x) = xj_{l+1}(x). \quad (12)$$

For each degree this transcendental equation has a countable infinity of positive roots $x_0 < x_1 < x_2 < \dots$, and the mode ${}_n T_l$ has eigenfrequency and period

$$\omega = \frac{\beta x_n}{b}, \quad \mathcal{T} = \frac{2\pi b}{\beta x_n}. \quad (13)$$

For $l = 2$, inserting $j_1(x) = \sin x / x^2 - \cos x / x$ and $j_2(x) = (3/x^3 - 1/x) \sin x - 3 \cos x / x^2$ into eq. (11) and clearing denominators gives the explicit form

$$\tan x = \frac{x(x^2 - 12)}{5x^2 - 12}, \quad (14)$$

which shows clearly that the roots must be found numerically. For large x we have $j_l(x) \approx \sin(x - l\pi/2)/x$, so eq. (11) becomes $\tan(x - l\pi/2) \approx x/2$ and successive roots are separated by approximately π : the overtones of a given degree become equally spaced in frequency, with spacing $\Delta f = \beta/2b$, which is the inverse of the two-way travel time of an S-wave across a diameter of the sphere. Finally, we note the special case $l = 1$, for which eq. (12) reduces to $j_2(kb) = 0$. This has the root $k = 0$, corresponding to $\omega = 0$ and, from the small-argument form of j_1 , to $W \propto r$. Since $r\mathbf{C}_{1m}$ is a rigid rotation (see Example 5), this is one of the trivial zero-frequency modes discussed in Lecture 22, and there is no fundamental toroidal mode of degree one.

(iii) *Numerical results.* The roots of eq. (11) were found by locating sign changes of $xj'_l(x) - j_l(x)$ on a fine grid and refining each by bisection, with j_l and j'_l evaluated using `scipy.special.spherical_jn`. With $b = 6371$ km and $\beta = 6.3$ km s⁻¹ we obtain the values in the table below. The shear wave speed chosen is a crude average over the mantle, whose values in PREM range from about 4.5 km s⁻¹ beneath the crust to about 7.3 km s⁻¹ at its base.

mode	$x_n = kb$	$f = \omega/2\pi$ (mHz)	period (min)
${}_0T_2$	2.5011	0.3936	42.3
${}_1T_2$	7.1360	1.1231	14.8
${}_2T_2$	10.5146	1.6548	10.1
${}_0T_3$	3.8647	0.6082	27.4
${}_1T_3$	8.4449	1.3291	12.5
${}_2T_3$	11.8817	1.8700	8.9

The predicted period of ${}_0T_2$ is 42.3 min, about 4% shorter than the observed 44.2 min ($f = 0.377$ mHz); a shear wave speed of 6.03 km s^{-1} would reproduce the observation exactly. Given how crude the model is, this agreement is better than we have any right to expect, and it is partly fortuitous. The Earth has a fluid outer core in which toroidal motion cannot be sustained, so the real mode is confined to the mantle, a shell occupying only the outer 45% of the radius; this stiffening of the problem raises the frequency. Against this, the low shear wave speeds of the upper mantle, where the fundamental mode has its largest displacements (Fig. 1), lower the frequency, and the two effects roughly compensate. For ${}_0T_3$ the model gives 27.4 min against an observed period of about 28.4 min. Fig. 1 shows the radial eigenfunctions of the three lowest $l = 2$ modes. The number of interior zeros of W equals the overtone number n , as the oscillation theorem for Sturm–Liouville problems guarantees, and the higher overtones oscillate increasingly rapidly with radius while the fundamental mode is smooth and largest at the surface.

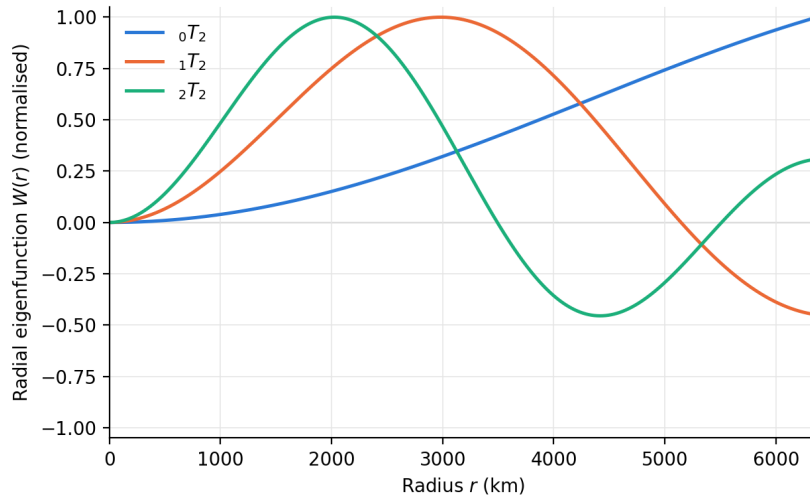


Fig. 1: Radial eigenfunctions $W(r) = j_2(x_n r/b)$ of the toroidal modes ${}_0T_2$, ${}_1T_2$ and ${}_2T_2$ of a homogeneous sphere with $b = 6371$ km, each normalised to unit maximum absolute value. The mode ${}_nT_2$ has n zeros in the interior, and all three satisfy the traction-free condition $\dot{W} = W/b$ at the surface.

This example illustrates how the abstract eigenvalue problem of the lecture becomes a concrete boundary value problem once spherical symmetry is used, and that even the simplest conceivable earth model predicts the gravest toroidal period to within a few per cent.

Example 2: The diagonal sum rule for a degree-one multiplet. For a multiplet with $l = 1$ the first-order splitting problem of the lecture is the 3×3 Hermitian eigenvalue problem $\sum_m M_{m'm} a_m = \delta\omega a_{m'}$, with $m, m' \in \{-1, 0, 1\}$ and

$$M_{m'm} = \frac{1}{2\omega_k} \left[-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle \right].$$

Suppose that, ordering the indices as $m = -1, 0, 1$, this splitting matrix is

$$M = \begin{pmatrix} -\Delta & \epsilon & 0 \\ \epsilon & 0 & \epsilon \\ 0 & \epsilon & \Delta \end{pmatrix},$$

with $\Delta > 0$ and ϵ real, the diagonal part representing splitting by rotation (see Example 3) and the off-diagonal part a small contribution from lateral heterogeneity.

- (i) Find the three singlet eigenfrequency perturbations and verify that the arithmetic mean of the split eigenfrequencies equals the degenerate eigenfrequency ω_k .
- (ii) Find the corresponding coefficient vectors a_m , verify that they are orthonormal, and explain what this implies for the zeroth-order eigenfunctions $\mathbf{s} = \sum_m a_m |km\rangle$.
- (iii) Explain why the conclusion of part (i) holds for any Hermitian splitting matrix with zero trace.

Solution. (i) The eigenvalues $\delta\omega$ are the roots of $\det(M - \delta\omega I) = 0$. Expanding along the first row,

$$\begin{aligned} \det(M - \delta\omega I) &= (-\Delta - \delta\omega) \left[-\delta\omega(\Delta - \delta\omega) - \epsilon^2 \right] - \epsilon [\epsilon(\Delta - \delta\omega)] \\ &= -\delta\omega (\delta\omega^2 - \Delta^2 - 2\epsilon^2), \end{aligned} \quad (15)$$

where the terms in $\Delta\epsilon^2$ cancel. The singlet perturbations are therefore

$$\delta\omega = -\Lambda, \quad 0, \quad +\Lambda, \quad \Lambda = \sqrt{\Delta^2 + 2\epsilon^2}, \quad (16)$$

and the split eigenfrequencies are $\omega_k - \Lambda$, ω_k and $\omega_k + \Lambda$, whose arithmetic mean is ω_k . Rotation alone ($\epsilon = 0$) gives three equally spaced lines with spacing Δ ; the heterogeneity leaves the central singlet where it is but pushes the outer two further apart, since $\Lambda > \Delta$. As an illustration, if $\Delta/2\pi = 3 \mu\text{Hz}$ and $\epsilon/2\pi = 2 \mu\text{Hz}$ then $\Lambda/2\pi = \sqrt{17} \mu\text{Hz} = 4.12 \mu\text{Hz}$.

(ii) For $\delta\omega = 0$ the equations $M\mathbf{a} = 0$ give $a_0 = (\Delta/\epsilon)a_{-1}$ and $a_1 = -a_{-1}$, while for $\delta\omega = \pm\Lambda$ the first and third rows give $a_0 = (\Delta \pm \Lambda)a_{-1}/\epsilon$ and $a_0 = (\Delta \mp \Lambda)a_1/\epsilon$. Using $\Lambda^2 - \Delta^2 = 2\epsilon^2$ to simplify, and normalising, we find

$$\mathbf{a}^{(0)} = \frac{1}{\Lambda} \begin{pmatrix} \epsilon \\ \Delta \\ -\epsilon \end{pmatrix}, \quad \mathbf{a}^{(\pm)} = \frac{1}{2\Lambda} \begin{pmatrix} \Lambda \mp \Delta \\ \pm 2\epsilon \\ \Lambda \pm \Delta \end{pmatrix}. \quad (17)$$

It is readily checked that $M\mathbf{a}^{(\pm)} = \pm\Lambda\mathbf{a}^{(\pm)}$; for example the middle row of $M\mathbf{a}^{(+)}$ is $\epsilon(\Lambda - \Delta) + \epsilon(\Lambda + \Delta) = 2\epsilon\Lambda$, which is Λ times the middle entry of $\mathbf{a}^{(+)}$. For the norms, $\|\mathbf{a}^{(0)}\|^2 = (\Delta^2 + 2\epsilon^2)/\Lambda^2 = 1$ and

$$\|\mathbf{a}^{(\pm)}\|^2 = \frac{(\Lambda \mp \Delta)^2 + 4\epsilon^2 + (\Lambda \pm \Delta)^2}{4\Lambda^2} = \frac{2\Lambda^2 + 2\Delta^2 + 4\epsilon^2}{4\Lambda^2} = 1, \quad (18)$$

while for the inner products

$$\begin{aligned}\mathbf{a}^{(+)} \cdot \mathbf{a}^{(-)} &= \frac{(\Lambda^2 - \Delta^2) - 4\epsilon^2 + (\Lambda^2 - \Delta^2)}{4\Lambda^2} = 0, \\ \mathbf{a}^{(0)} \cdot \mathbf{a}^{(\pm)} &= \frac{\epsilon(\Lambda \mp \Delta) \pm 2\epsilon\Delta - \epsilon(\Lambda \pm \Delta)}{2\Lambda^2} = 0.\end{aligned}\quad (19)$$

The three coefficient vectors are thus orthonormal, as the eigenvectors of a Hermitian matrix with distinct eigenvalues must be. Because the singlets satisfy $\langle km|P_0|km'\rangle = \delta_{mm'}$, the zeroth-order eigenfunctions $\mathbf{s}^{(j)} = \sum_m a_m^{(j)} |km\rangle$ obey

$$\langle \mathbf{s}^{(j)} | P_0 | \mathbf{s}^{(j')} \rangle = \sum_m a_m^{(j)*} a_m^{(j')} = \delta_{jj'}, \quad (20)$$

so they are orthonormal with respect to the kinetic energy form, exactly as Lecture 22 requires of eigenfunctions with distinct eigenfrequencies. Note that for $\epsilon \neq 0$ each of these eigenfunctions is a mixture of all three singlets; the heterogeneity has selected a particular orthonormal basis of the degenerate eigenspace, and it is this basis, not the original singlets, that the perturbed eigenfunctions approach as the perturbation is switched off. As $\epsilon \rightarrow 0$ we recover $\mathbf{a}^{(\pm)} \rightarrow (0, 0, 1)$ and $(1, 0, 0)$, the singlets themselves.

(iii) For any Hermitian matrix we can write $M = U \text{diag}(\delta\omega_1, \dots, \delta\omega_{2l+1}) U^\dagger$ with U unitary, and the cyclic property of the trace gives $\text{tr } M = \text{tr}(\text{diag}(\delta\omega_j) U^\dagger U) = \sum_j \delta\omega_j$. Equivalently, the coefficient of $\delta\omega^{2l}$ in the characteristic polynomial is $-\text{tr } M$ up to sign, as eq. (15) illustrates. Hence if $\text{tr } M = 0$ the singlet perturbations sum to zero, and the mean of the $2l + 1$ split eigenfrequencies is ω_k . The content of the diagonal sum rule of the lecture is the statement, following from the selection rules for the matrix elements, that the trace does vanish for splitting by rotation and by lateral heterogeneity, so that averaging the observed singlet frequencies recovers the eigenfrequency of the spherically averaged Earth.

Example 3: Rotational splitting is linear in m . Within a multiplet $|km\rangle$, $-l \leq m \leq l$, of a spherically symmetric earth model it can be shown, using the conventions of the lecture, that the matrix elements of the Coriolis form are

$$\langle km' | W | km \rangle = -2i\Omega\beta_k m \delta_{mm'},$$

where Ω is the angular rate of rotation and β_k is a dimensionless constant, the **splitting parameter**, that depends on the radial eigenfunctions of the multiplet. Take this result as given. (i) Check that it is consistent with the anti-self-adjointness of the Coriolis form. (ii) Neglecting the perturbations P_1 and H_1 (which for a rotating earth model describe the ellipticity of figure and the centrifugal potential), solve the first-order Hermitian eigenvalue problem of the lecture, and show that $\delta\omega_m = m\Omega\beta_k$, so that the multiplet splits into $2l + 1$ equally spaced lines. (iii) For ${}_0S_2$, with degenerate frequency $f_k = 0.3094$ mHz and $\beta_k \approx 0.40$, give the spacing of adjacent singlets and the total width of the multiplet in μHz , and comment on the length of record required to resolve the singlets.

Solution. (i) In Lecture 22 we showed that $\langle \mathbf{w} | W | \mathbf{u} \rangle = -\langle \mathbf{u} | W | \mathbf{w} \rangle^*$. Applied to the singlets this requires $\langle km' | W | km \rangle = -\langle km | W | km' \rangle^*$. With the stated matrix elements the right-hand side is $-(-2i\Omega\beta_k m' \delta_{m'm})^* = -2i\Omega\beta_k m' \delta_{mm'}$, which equals the

left-hand side because the Kronecker delta forces $m' = m$. In particular the diagonal elements $\langle km|W|km\rangle$ are purely imaginary, as anti-self-adjointness demands. They are also odd in m : the singlets $\pm m$ correspond to patterns $e^{i(\pm m\varphi + \omega t)}$ travelling in opposite directions around the rotation axis, and the Coriolis force acts on them with opposite sign.

(ii) Dropping P_1 and H_1 , the first-order eigenvalue problem of the lecture reduces to

$$\frac{1}{2\omega_k} \sum_{m=-l}^l i\omega_k \langle km'|W|km\rangle a_m = \delta\omega a_{m'}. \quad (21)$$

Inserting the matrix elements, the sum collapses to the single term $m = m'$, and

$$\frac{i}{2} (-2i\Omega\beta_k m') a_{m'} = \Omega\beta_k m' a_{m'} = \delta\omega a_{m'}, \quad -l \leq m' \leq l. \quad (22)$$

The splitting matrix is thus already diagonal, $M = \Omega\beta_k \text{diag}(-l, \dots, l)$, with real entries, as the diagonal of a Hermitian matrix must be. Its eigenvalues and eigenvectors can be read off: for each m ,

$$\delta\omega_m = m\Omega\beta_k, \quad a_{m'} = \delta_{m'm}, \quad (23)$$

so the zeroth-order eigenfunctions are the singlets $|km\rangle$ themselves, and the perturbed eigenfrequencies are

$$\omega_m = \omega_k + m\Omega\beta_k, \quad m = -l, \dots, l. \quad (24)$$

The multiplet splits into $2l + 1$ lines, equally spaced by $\Omega\beta_k$ and placed symmetrically about ω_k , which is the direct analogue of the normal Zeeman effect, $\delta E = m\mu_B B$, for an atom in a magnetic field. The shifts sum to $\Omega\beta_k \sum_m m = 0$, in accordance with the diagonal sum rule. The result holds to first order in Ω ; the neglected terms P_1 and H_1 add a contribution that is even in m and so slightly breaks the equal spacing, but for the gravest modes it is smaller than the Coriolis term.

For toroidal modes the splitting parameter can be found explicitly. With $|km\rangle = W\mathbf{C}_{lm}$ and $\mathbf{\Omega} = \Omega\hat{\mathbf{z}}$ the definition of the Coriolis form gives

$$\langle km|W|km\rangle = 2\Omega \int_0^b \rho W^2 r^2 dr \int_{S^2} \mathbf{C}_{lm}^* \cdot (\hat{\mathbf{z}} \times \mathbf{C}_{lm}) dS, \quad (25)$$

and a short calculation with the explicit components of $\mathbf{C}_{lm}$, which we omit, shows that the surface integral equals $-im$. With the normalisation $\zeta^2 \int_0^b \rho W^2 r^2 dr = 1$ from the lecture this gives $\langle km|W|km\rangle = -2i\Omega m / \zeta^2$, so that $\beta_k = 1/l(l+1)$ exactly for every toroidal multiplet. For ${}_0T_2$ the singlets are therefore spaced by $\Omega/(12\pi) = 1.93 \mu\text{Hz}$.

(iii) Using the sidereal day, $\Omega = 2\pi/86164 \text{ s} = 7.292 \times 10^{-5} \text{ s}^{-1}$, or $\Omega/2\pi = 11.61 \mu\text{Hz}$. For ${}_0S_2$ with $\beta_k = 0.40$ the spacing between adjacent singlets is

$$\delta f = \frac{\beta_k \Omega}{2\pi} = 4.64 \mu\text{Hz}, \quad (26)$$

and the five singlets span a total width of $2l\delta f = 18.6 \mu\text{Hz}$, which is 1.5% of f_k per singlet spacing. This agrees well with the observed singlet frequencies of ${}_0S_2$, which are spaced by about $4.6 \mu\text{Hz}$. To resolve two spectral lines separated by δf we saw in

the worked examples for Lecture 22 that the record length T must satisfy $T \gtrsim 1/\delta f$, which here is 2.2×10^5 s, or about 60 hours. The intrinsic width of each line due to attenuation is $f_k/Q \approx 0.6 \mu\text{Hz}$ for $Q \approx 510$, an order of magnitude smaller than the splitting, and the corresponding decay time $Q/\pi f_k \approx 6$ days comfortably exceeds the required record length. The rotational splitting of ${}_0S_2$ can therefore be resolved from a single seismogram several days long after a great earthquake, and indeed it was such a resolved splitting that motivated the first theoretical treatments of the problem.

Example 4: Temperature or composition? Write $\delta \ln \beta = \delta \beta / \beta$, $\delta \ln \gamma = \delta \gamma / \gamma$ and $\delta \ln \rho = \delta \rho / \rho$ for the fractional perturbations of shear wave speed, bulk sound speed and density relative to the spherical average at some depth in the lower mantle. Take the following illustrative values of the partial derivatives, of the magnitudes suggested by mineral physics. With respect to temperature, $\partial \ln \beta / \partial T = -1.0 \times 10^{-4} \text{ K}^{-1}$, $\partial \ln \gamma / \partial T = -3.0 \times 10^{-5} \text{ K}^{-1}$ and $\partial \ln \rho / \partial T = -1.5 \times 10^{-5} \text{ K}^{-1}$. With respect to a compositional variable X , taken to be the molar iron fraction $\text{Fe}/(\text{Mg} + \text{Fe})$, $\partial \ln \beta / \partial X = -0.30$, $\partial \ln \gamma / \partial X = +0.10$ and $\partial \ln \rho / \partial X = +0.30$. (i) Show that for purely thermal anomalies $\delta \ln \gamma / \delta \ln \beta > 0$, and evaluate the ratio. (ii) Tomography suggests that within an LLSVP $\delta \ln \beta \approx -2\%$ and $\delta \ln \gamma \approx +0.5\%$. Show that these values cannot be explained by temperature alone, and solve for the temperature and compositional anomalies δT and δX that explain both. (iii) Compute the implied density anomaly for (a) a purely thermal interpretation of the shear wave anomaly alone and (b) the joint solution of part (ii), and comment.

Solution. (i) For a purely thermal anomaly the lecture's linearised relations give $\delta \ln \beta = (\partial \ln \beta / \partial T) \delta T$ and $\delta \ln \gamma = (\partial \ln \gamma / \partial T) \delta T$, so that

$$\frac{\delta \ln \gamma}{\delta \ln \beta} = \frac{\partial \ln \gamma / \partial T}{\partial \ln \beta / \partial T} = \frac{-3.0 \times 10^{-5}}{-1.0 \times 10^{-4}} = 0.3. \quad (27)$$

The ratio is positive because both derivatives are negative: heating a crystalline solid reduces its elastic moduli more rapidly than thermal expansion reduces its density, so all seismic wave speeds fall. Thermal anomalies therefore produce positively correlated shear and bulk sound speed anomalies, with the bulk sound speed the less sensitive of the two.

(ii) If the shear wave anomaly were thermal we would need $\delta T = \delta \ln \beta / (\partial \ln \beta / \partial T)$, which is $(-0.02) / (-1.0 \times 10^{-4}) = 200 \text{ K}$, and eq. (27) would then predict $\delta \ln \gamma = 0.3 \times (-2\%) = -0.6\%$. The observed value of $+0.5\%$ has the opposite sign, so no choice of δT alone can fit both observations. Allowing for composition, the lecture's relations become the linear system

$$\begin{pmatrix} -1.0 \times 10^{-4} & -0.30 \\ -3.0 \times 10^{-5} & 0.10 \end{pmatrix} \begin{pmatrix} \delta T \\ \delta X \end{pmatrix} = \begin{pmatrix} -0.020 \\ 0.005 \end{pmatrix}, \quad (28)$$

in which δT is in kelvin. The determinant is $(-1.0 \times 10^{-4})(0.10) - (-0.30)(-3.0 \times 10^{-5}) = -1.9 \times 10^{-5}$, and Cramer's rule gives

$$\begin{aligned} \delta T &= \frac{(0.10)(-0.020) - (-0.30)(0.005)}{-1.9 \times 10^{-5}} = 26 \text{ K}, \\ \delta X &= \frac{(-1.0 \times 10^{-4})(0.005) - (-3.0 \times 10^{-5})(-0.020)}{-1.9 \times 10^{-5}} = 0.058. \end{aligned} \quad (29)$$

With these numbers the LLSVP is barely warmer than its surroundings and is enriched in iron by nearly six mole per cent; almost the whole of the shear wave anomaly is compositional.

(iii) For interpretation (a), $\delta T = 200$ K and

$$\delta \ln \rho = \frac{\partial \ln \rho}{\partial T} \delta T = -1.5 \times 10^{-5} \times 200 = -0.3\%, \quad (30)$$

so the LLSVP would be lighter than the ambient mantle and buoyant: the “super plume” picture. For interpretation (b),

$$\delta \ln \rho = \frac{\partial \ln \rho}{\partial T} \delta T + \frac{\partial \ln \rho}{\partial X} \delta X = -1.5 \times 10^{-5} \times 26 + 0.30 \times 0.058 = +1.7\%, \quad (31)$$

so the same region would be denser than its surroundings and would tend to remain at the base of the mantle. The two interpretations thus differ not merely in the size but in the sign of the density anomaly, and hence in the dynamical role of the LLSVPs. The joint solution is, moreover, sensitive to derivatives that are poorly known: repeating the calculation with $\partial \ln \gamma / \partial X = +0.05$ instead of $+0.10$ gives $\delta T = -36$ K, $\delta X = 0.079$ and $\delta \ln \rho = +2.4\%$, so even the sign of the temperature anomaly is not robust, although the density excess is. What this example shows is that wave speeds alone, however well resolved, cannot determine whether LLSVPs are light or dense once compositional variations are admitted, and this is why direct constraints on lateral density variations from free oscillations and tides are of such interest.

Example 5: Vector spherical harmonics of degree one. The spherical harmonic $Y_{10} = \sqrt{3/4\pi} \cos \theta$ is real and normalised. (i) Compute $\mathbf{A}_{10}$, $\mathbf{B}_{10}$ and $\mathbf{C}_{10}$ explicitly, and verify that $\int_{S^2} \mathbf{B}_{10}^* \cdot \mathbf{C}_{10} dS = 0$ and $\int_{S^2} \|\mathbf{B}_{10}\|^2 dS = \int_{S^2} \|\mathbf{C}_{10}\|^2 dS = l(l+1) = 2$. (ii) Show that $\mathbf{A}_{10} + \mathbf{B}_{10}$ is a constant vector and that $r\mathbf{C}_{10}$ is a rigid rotation, and interpret these results in terms of the trivial modes of Lecture 22.

Solution. (i) Since Y_{10} does not depend on φ , the tangential gradient is $\nabla_1 Y_{10} = \hat{\theta} \partial Y_{10} / \partial \theta$, and from the definitions in the lecture

$$\begin{aligned} \mathbf{A}_{10} &= \sqrt{\frac{3}{4\pi}} \cos \theta \hat{\mathbf{r}}, & \mathbf{B}_{10} &= \nabla_1 Y_{10} = -\sqrt{\frac{3}{4\pi}} \sin \theta \hat{\theta}, \\ \mathbf{C}_{10} &= -\hat{\mathbf{r}} \times \nabla_1 Y_{10} = \sqrt{\frac{3}{4\pi}} \sin \theta \hat{\varphi}, \end{aligned} \quad (32)$$

where we have used $\hat{\mathbf{r}} \times \hat{\theta} = \hat{\varphi}$. Because $\hat{\theta}$ and $\hat{\varphi}$ are orthogonal at every point, $\mathbf{B}_{10}^* \cdot \mathbf{C}_{10} = 0$ pointwise and its integral vanishes trivially. With $dS = \sin \theta d\theta d\varphi$,

$$\int_{S^2} \|\mathbf{B}_{10}\|^2 dS = \frac{3}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi \sin^3 \theta d\theta = \frac{3}{4\pi} \times 2\pi \times \frac{4}{3} = 2, \quad (33)$$

using $\int_0^\pi \sin^3 \theta d\theta = \int_{-1}^1 (1-c^2) dc = 4/3$, and the same integrand gives $\int \|\mathbf{C}_{10}\|^2 dS = 2$. Both agree with $l(l+1) = 2$. That the answer is $l(l+1)$ in general follows by integrating by parts on the sphere, $\int \nabla_1 Y_{lm}^* \cdot \nabla_1 Y_{lm} dS = -\int Y_{lm}^* \nabla_1^2 Y_{lm} dS = l(l+1)$, using the defining eigenvalue relation for Y_{lm} ; the present calculation is just this identity made explicit for one harmonic.

(ii) Recalling that $\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}}$, eq. (32) gives

$$\mathbf{A}_{10} + \mathbf{B}_{10} = \sqrt{\frac{3}{4\pi}} \hat{\mathbf{z}}, \quad (34)$$

a constant vector. Similarly $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \sin \theta \hat{\boldsymbol{\phi}}$, so that

$$r\mathbf{C}_{10} = \sqrt{\frac{3}{4\pi}} \hat{\mathbf{z}} \times \mathbf{x}, \quad (35)$$

which is the displacement produced by an infinitesimal rigid rotation about the z -axis. In the language of the lecture, the spheroidal displacement $U\mathbf{A}_{10} + V\mathbf{B}_{10}$ with $U = V = \text{const}$ is a rigid translation along $\hat{\mathbf{z}}$, and the toroidal displacement $W\mathbf{C}_{10}$ with $W \propto r$ is a rigid rotation about $\hat{\mathbf{z}}$; the harmonics with $m = \pm 1$ supply the translations and rotations about the x - and y -axes. These are the six zero-frequency trivial modes of Lecture 22, and we see that they are all of degree one. This is consistent with Example 1, where the $l = 1$ frequency equation was found to have the root $\omega = 0$ with $W \propto r$, and it explains why the mode catalogues of the Earth contain no ${}_0T_1$ and no ${}_0S_1$. A gravitating earth model does have a non-trivial degree-one spheroidal mode, ${}_1S_1$, in which the inner core translates against the rest of the Earth, but it is not a rigid motion.